Benchmarking Hybrid Graph Transactional/Analytical Processing and Elastic Scale-Out in Open-Source Distributed Graph Databases

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Abstract

Today, graph intensive applications that we frequently encounter such as fraud analysis, real time recommendation systems and knowledge graph construction require database systems that can manage highly dynamic graph data and also operate under different workload intensities. Among these workloads, there are operations called multi hop traversals, pattern matching processes and high speed update requirements.

When transaction mechanisms are activated in such environments, the problem is named as Hybrid Graph Transactional/Analytical Processing (HGTAP) [1, 2]. HGTAP requires Graph Transactional Processing (GTP) and Graph Analytical Processing (GAP) processes to work simultaneously and under strict consistency constraints.

In recent years, several studies proposed hybrid benchmarking frameworks in order to evaluate the performance of HTAP systems [3, 4]. However, there is no recent cross-system benchmarking study that systematically compares open-source distributed graph databases under concurrent HGTAP workloads together with elastic scale-out scenarios. Most existing works either evaluate newly proposed research prototypes or conduct experiments under fixed cluster configurations. In addition, they do not analyze system behavior during dynamic node addition while hybrid transactional and analytical workloads are actively running.

This study proposes a comprehensive benchmarking work on five popular open source distributed graph database systems: NebulaGraph [5], JanusGraph [6], HugeGraph [7], Dgraph [8] and ArangoDB [9]. These systems represent different architectural approaches

such as LSM tree based native graph storage, pluggable wide column backends, predicate based sharding strategies and multi model document storage.

In the proposed study, LDBC Financial Benchmark (FinBench) [10], its extended version FinBench-X [11] and the OLXP workload model defined in HyBench [3] will be used. The evaluation will be conducted based on transactional throughput (TPS), analytical throughput (QPS), hybrid throughput (XPS), latency statistics, workload interference between different operations and data freshness.

In addition to this, it is expected that systems using LSM-tree based approaches may show higher performance under write intensive workloads, while systems with multi-model structure may exhibit different performance behaviour in concurrent hybrid processing scenarios.

How This Paper Will Fulfill the Requirements

Project Criteria Fulfillment

Choose a data management/databases problem for which several algorithms/approaches are published. Benchmark different approaches on various public datasets from different dimensions such as performance, accuracy, etc. There are at least 5 different approaches proposed to solve the problem in the literature. These approaches should ideally either have public GitHub repositories or be easy to implement within the semester timeframe. There should not be a benchmarking paper published within the last 5 years on the same topic covering the intended methods. Or, if such a paper exists, the proposed work should clearly differentiate itself by filling important gaps.

How This Proposal Satisfies the Criteria

- **Clearly defined problem.** The main problem of this project is benchmarking distributed graph databases in terms of concurrent HGTAP performance and how flexible they are during dynamic scale-out. Shortly, I compare how these systems behave when transactional and analytical workloads run at same time and when new nodes are added into system.

- **At least five different approaches.** In this study five different graph database systems with different architectures are selected: NebulaGraph, JanusGraph, HugeGraph, Dgraph and ArangoDB. These systems have different design choices and this makes comparison more meaningful.

- **Open-source systems.** All selected database systems are open source and their source codes are available on GitHub, so they are accesible for experimental evaluation.

- **Open-source datasets.** The datasets used in this study are also open source. FinBench, its extended version FinBench-X and OLXP workload model defined in HyBench are publicly available and commonly used in benchmarking studies.

- **Multi-dimensional evaluation.** As mentioned in abstract the systems will not be evaluated from only one aspect. They will be compared based on transactional throughput (TPS), analytical throughput (QPS), hybrid throughput (XPS), latency values workload interference and data freshness. Also their behaviour during dynamic scale-out will be analysed.

- **Differentiation from existing literature.** Studies such as HyBench [3], BACH [2] and Galaxybase [4] work on hybrid processing but they mostly focus on their own newly developed systems or fixed cluster environments. In this project instead of proposing a new system, I compare existing open source databases and analyse them under concurrent hybrid workloads and dynamic node addition under active load. This cross system comparison makes the study different from others.

Note

This proposal is written without including any numerical results or specific claims, since there are currently no benchmark results available. As it can be expected, it is tentative and it may change significantly during the progress of the study.

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